

Observer-Limited Geometric Cognition

The identification problem, and what has to hold for the theory
to say anything a flexible embedding does not

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Every statement is labeled *proved*, *defined*, *posited*, or *open*.

Abstract

Observer-Limited Geometric Cognition holds that an agent represents a task under finite resolution, evaluates represented consequences in a geometry of its own, and selects behavior by bounded search. Preference is then a derived object rather than a primitive. The formal core of this picture is not new. Geometric Evaluation Theory supplies the evaluation object and proves that a rank budget coarsens distinctions, that the induced preference is a semiorder with threshold equal to the budget, and that a budget change reverses preference between fixed actions. Observation Theory supplies the prior layer, in which what a consumer can read is a property of the pair and not of the system. This paper argues that the open problem is neither geometry nor dynamics but identification. A representation, a metric, a reference, an admissible set and a search policy together form a family expressive enough to fit any finite choice record, so a theory that fits every record predicts nothing. We state that as a proposition rather than as a worry, and give the conditions under which the account carries content. The conditions are modular. Each of the observer's four components must be identified on evidence disjoint from the choices the theory is asked to predict. A cold reread of the first phase's registration then shows that this condition is not sufficient. Where the class a pair is assigned to is itself read off the fitted model, the graded contrast is an algebraic identity in the evaluator's accuracy and measures nothing else, so content is not monotone in identification accuracy and we can bound neither end. A rank budget carries a further condition that modular identification does not imply, since the retained subspace is identified only where the workload moment has a spectral gap, and a design that places the ideal at the centre of its own consequence distribution has no gap and recovers a direction fixed by sampling noise. What survives is that identification is the step that fails in practice, since two of three candidate evaluators could not be calibrated at all. We restate the five predictions of the programme as consequences of those conditions, restate the seven falsifiers as conditions on the record rather than on the author's judgment, and give a phased programme in which each phase retires exactly one degree of freedom. The theory's own epistemic status is that the evaluation layer is proved and measured on artificial evaluators, the observer layer is measured in other domains, and the cognitive synthesis is posited.

Keywords: geometric cognition, observer-relative representation, resolution budget, preference reversal, identification, bounded search

1 What this paper claims and does not claim

The thesis behind this paper is that intelligent agents act by searching over observer-relative, resource-limited geometries of possible states. This paper does not argue for that thesis. It asks the prior question of what would have to be true for the thesis to be testable at all, and answers it.

Three things are already established elsewhere and are used here without reproof. Geometric Evaluation Theory (GET) proves that unbudgeted distinctions form an equivalence, that a rank budget coarsens it, that the induced preference is a weak order at zero threshold and a semiorder at positive threshold with the threshold fixed by the budget, and that a rank budget change reverses preference between fixed actions. Observation Theory (OT) establishes in several domains that what a consumer can recover is a property of the pair formed by the system and the consumer rather than of the system alone. A domain exposition of geometric cognition exists and reports measurements on artificial agents.

One thing is not established, and this paper is about it. The domain exposition measured cognition through a representation that it also chose. Its admission filter recorded the resulting circularity and refused the instrument, on the ground that no structure label independent of the embedding was available. That refusal is correct and it is the starting point here. A geometric account of cognition whose representation is fitted to the same choices it explains is not a weak theory. It is not yet a theory.

2 The evaluation object and the observer above it

Definition 1 (Evaluation object, after GET). An evaluation object is a tuple

$$E = (X, A, C, G, B, K)$$

with states X carrying a belief measure μ , actions A , a consequence map $C: A \times X \rightarrow Y$, a pointed geometry G consisting of a Riemannian metric on Y and an ideal t , a resolution budget B , and an admissible set K that may depend on the state or on the menu.

The commitment that makes the object non-trivial is the split. The world supplies (X, A, C) and the evaluator supplies (G, B, K) . Without it an evaluator free to choose C after seeing the desired ordering represents any finite order by construction, and representability is a tautology. With it, representation and identification become questions with answers.

Evaluation is by expected squared geodesic distance from the ideal, and cost is that distance squared,

$$d(a) = [\mathbb{E}_\mu d_G(C(a, X), t)^2]^{1/2}, \quad c(a) = d(a)^2.$$

A metric alone orders nothing. Ordering appears only relative to a reference, which separates the question of which differences the agent is sensitive to, carried by G , from the question of where those differences are measured from, carried by t .

Under a rank budget k the evaluator retains the top- k eigenspace of the workload-weighted second moment,

$$M_\omega = \mathbb{E}_\omega [G^{1/2}(C - t)(C - t)^\top G^{1/2}], \quad d_k(a) = [\mathbb{E}_\mu \|\Pi_k G^{1/2}(C(a, X) - t)\|^2]^{1/2}.$$

Two consequences that differ only in discarded directions are then indistinguishable to the evaluator. This is the structural claim that separates the account from a noise model. A budget does not blur a complete representation. It removes directions.

Definition 2 (Observer layer). A cognitive observer adds a representation map $\Phi: W \rightarrow Y$ from the world’s task structure W to the consequence space on which the evaluation object is defined, and a search policy π over represented states. The observer is the quadruple (Φ, G, B, K) together with π . The world retains W and the action set.

Definition 2 is where OT enters. Φ is a read operator, and OT’s finding is that its content is a property of the pair and not of the world. The cognitive reading is that two agents facing the same task can face different operational problems, and that this happens before anything that could be called a preference.

3 The vacuity problem, stated as a proposition

The programme’s own statement of limits names flexibility as the principal threat. We state it more sharply, because a threat that is described in prose can be acknowledged indefinitely without being retired.

Proposition 3 (Vacuity without modular identification). *Let a finite choice record consist of menus $M_1, \dots, M_n$ drawn from a finite alternative set and a selection from each. If the observer’s four components (Φ, G, B, K) are all fitted to that record, then for any record there exists an observer reproducing it exactly. The account therefore excludes no finite choice record, and no observation of choice alone can disconfirm it.*

Status: proved, elementary, and not new. Admissibility alone suffices. Taking K menu-dependent and equal to the singleton actually chosen reproduces any record whatever, for any Φ , G and B .

The proposition belongs to a known family and is restated here only because the components are ours. Kalai, Rubinstein and Spiegel show that rationalization by multiple rationales admits every choice function, so the model measures degrees of rationality rather than rationality [7]. Under endogenous menu assignment every dataset of menu frequencies and conditional choice probabilities is rationalizable by some random-utility distribution, and the classical axioms lose their content entirely. The contrast with Afriat is the instructive part [6]. On budget data, rationalizability by a concave monotone utility has real content, because GARP can fail. What removes the content in our setting is menu-dependent admissibility, which is exactly the component GET keeps separate from cost because it can violate the weak axiom. The separation that makes K theoretically interesting is the same separation that makes the account untestable when K is free.

The proposition’s interest is therefore not mathematical. It is that vacuity arrives through the weakest of the four components, so no argument about the plausibility of a metric or the realism of a budget repairs it.

Proposition 3 has the shape of a defect this programme has met before in another arm. A conformance property evaluated over an empty dependency graph passes perfectly and certifies nothing, and the repair there was to require that the structure the property quantifies over be non-vacuous before the property may be reported. The repair here is the same in form. The components of the observer must be identified on evidence that the prediction does not use.

Definition 4 (Modular identification). An OLGC account of a choice record R is *modularly identified* when each of Φ , G , B and π is estimated from evidence disjoint from R , and K is either fixed in advance or derived from a stated constraint that is not a function of the selections in R .

Proposition 5 (Content under modular identification). *Under Definition 4 with Φ , G and B fixed before R is opened, the induced order on any menu is determined, and the account therefore excludes every record that differs from the induced one by more than the semiorder’s own indifference band.*

Status: proved, given GET’s Theorem 2. The band is ε_B and is fixed by the budget rather than fitted, which is the property that makes the exclusion non-empty. The proposition is the reason the programme’s experimental designs insist on separate calibration blocks. It converts a methodological preference into a condition for the theory to have content.

A second repair exists and does not require external identification. Caplin and Dean give a revealed-preference test for optimal acquisition of costly information whose no-improving-attention-cycles condition is necessary and sufficient for a finite dataset to be consistent with the model, and they recover information costs from choice data alone [8]. Related work identifies consideration sets from choice in the same spirit [9, 10]. That line and this one answer the same threat differently. Theirs constrains the model class until choice data discriminate within it. Ours holds the model class and constrains the evidence. Which is preferable is an empirical question about how much external identification costs, and it is not settled here. A budget recovered by a cycle condition and a budget manipulated experimentally are different objects, and P1 below is a test of the second.

Open 6. Proposition 5 assumes full prior identification. The realistic case is partial, where one component is estimated jointly with the prediction. The degrees of freedom retired by identifying Φ alone, or B alone, are not characterized, and no bound is known on how much a single free component can absorb. Until such a bound exists, an account that identifies three of four components is not known to be more falsifiable than one that identifies none.

Identification can also be too good

Open 6 is one end of a problem with two ends. Under-identified, a free component absorbs the prediction and nothing is excluded. The other end appeared in the Phase I pilot and was not anticipated when Proposition 5 was written.

The Phase I design identifies G and t from a calibration block of ordinal comparisons, then admits a pair into the trading class precisely when the fitted metric says the pair’s rank- k order disagrees with its full-budget order, and into the within-subspace class precisely when it says the order is preserved. Reversal is predicted for the first class and not for the second. The class label is therefore the prediction, and the consequence is not approximate. Write p for the probability that a graded verdict agrees with the fitted metric. A trading pair reverses when the evaluator agrees with the fit at both budgets or errs at both, and a within-subspace pair reverses when it errs at exactly one, so

$$R_T = p^2 + (1 - p)^2, \quad R_W = 2p(1 - p), \quad R_T - R_W = (2p - 1)^2.$$

The contrast the design grades is a function of the evaluator’s per-comparison accuracy and of nothing else. On the pinned evaluator the observed contrast of 0.968 against 0.000 implies

$p = 0.992$, against a measured held-out accuracy of 0.985. No part of the separation is left for the budget manipulation to explain, and the pass bars are the same quantity restated: the registered margin is met exactly when p exceeds 0.848.

This is not an artifact of one design. Any test of the form *identify the components, then predict a choice from them* inherits the difficulty, because the prediction is a function of the identified components and its informativeness falls as the identification error falls toward zero. Content is therefore not monotone in identification accuracy. It is low at both ends and the maximum is somewhere in between, and we can locate neither end.

The same reread found a second condition, on B rather than on G , which Definition 4 does not state and should. A rank budget is a choice of top- k eigenspace of the workload moment $M = \mathbb{E}[G^{1/2}(C - t)(C - t)^\top G^{1/2}]$, and that eigenspace is identified only if M has a spectral gap at k . The Phase I design drew consequences uniformly on a cube and placed the ideal at the cube’s centroid, which makes M isotropic analytically, at $(\text{hi} - \text{lo})^2/12$ times the identity. There is then no gap and no preferred eigenspace, and the retained subspace is fixed by which finitely many calibration pairs were drawn. On an evaluator that is exactly Euclidean, with no response error whatever, two independent calibration draws select retained directions a median of 56 degrees apart, against the 60 degrees expected between two uniformly random directions in three-space. Under that design a budget manipulation is a projection onto an arbitrary plane, and a confirmation of P2 would report a property of orthogonal projection. Identifying Φ and G on disjoint evidence does not help, because the defect is in the workload distribution the moment is taken over.

Definition 7 (Budget identification). A rank budget $B = k$ is identified for an evaluation object only if the workload moment M has a spectral gap at k , and an experiment that manipulates B must exhibit that gap against the sampling distribution of M at the realized sample size rather than assume it.

The condition is easy to satisfy and easy to omit, and omitting it costs the whole experiment. It also shows that the anti-vacuity check the programme inherited from its sibling arm is not sufficient here. That check requires a non-trivial share of the trace to be discarded, and isotropy discards the largest such share while distinguishing nothing.

Open 8. Proposition 5 treats prior identification as a good whose quantity is to be maximized. The pilot shows it is not. No bound is known on how accurate an identification may be before the prediction it licenses becomes a consequence of the fit, and no measure separates the residual content from the entailed part. Until one exists, a design of this shape should register identification accuracy as an interval and not as a floor, and should expect the informative work to be done by the identification step rather than by the prediction it feeds.

The consolation is that the identification step is where the failures actually occurred. Of three candidate evaluators put through the Phase I calibration, two could not be calibrated at all. One ordered by presentation position at a first-position rate of 0.8175, with an agreement rate across the two orders that matched what pure position preference predicts and nothing that a metric would explain. A second recovered distances at $r^2 = 0.766$ and still ordered by position at 0.700. Only the third compared by content. An evaluator can fail to be a quadratic evaluation object, and most of the ones we tried did, so the step that carries the content is the step that can fail.

4 What is proved, what is measured, and what is posited

The programme’s statements do not carry a uniform status, and a foundational paper that presents them at one strength misrepresents all of them.

Table 1: Status of the programme’s statements. The evaluation layer is proved, and measured on artificial evaluators except where a row says otherwise. Every statement specific to *cognition* sits in the lower block and carries no sealed measurement.

Statement	Status	Basis
Rank budget coarsens distinctions	proved	GET Theorem 1, machine-checked
Preference is a semiorder with threshold the budget	proved	GET Theorem 2, machine-checked
Budget change reverses preference between fixed actions	proved (existence)	GET Theorem 5, machine-checked
Threshold tracks budget in an evaluator	demonstrated	GET gate G3, sealed. A language-model scorer’s indifference threshold is 0.87 to 0.88 of the rendering step, and every pair separated by 20 is ordered alike in every cell
Hull law in a population	demonstrated as a tendency	GET gate G2, sealed, ANES 1972. Half of 489 testable respondents violate at least once against a shuffle null of 66 percent. It fails as a deterministic statement
An evaluator can fail to be a quadratic evaluation object	measured, not sealed	Phase I instrument search [4]. Two of three candidate language-model evaluators could not be calibrated, ordering by presentation position at 0.8175 and 0.700. The third calibrated and held out at 0.985, which is a maximum selected over the three and should be read as one
Representation is prior to preference	posited for cognition	measured in other domains by OT, not in a cognitive observer
Human cognitive budgets exist and are manipulable	posited	no sealed measurement
Bounded search is the right dynamical abstraction	posited	alternatives in §6 are not excluded
Whitney stratification of cognitive state space	posited	no measurement distinguishes it from a steep smooth family
Cross-architecture universality	posited	no collapse has been attempted

The four rows at the bottom are the content of the synthesis. They are also the rows with no sealed measurement behind them. That is the position of the programme at draft 0.1 and it is stated here rather than in a closing section.

5 Predictions, as consequences of the identification conditions

The programme states five predictions. Each is restated here as a claim that becomes testable only under Definition 4, with the component whose independent identification it requires named beside it.

P1, the threshold-budget law. Holding G and t fixed and manipulating B , the just-

noticeable comparison threshold tracks the budget. Well separated pairs keep their order. Requires independent identification of G and t . A systematic reordering of well separated pairs under a pure resolution manipulation contradicts the model.

P2, direction-specific rank reversal. Reversals concentrate on pairs trading a retained direction against a newly unresolved one. Pairs whose difference lies in the retained subspace keep their order. Requires independent identification of Φ , since the retained subspace is a property of Φ and B together. This is the prediction that separates the account from a claim that load adds noise. Definition 7 and Open 8 bound what a confirmation of it is worth. The retained subspace must be identified before it can be manipulated, and where the class assignment is made by the same fitted metric that predicts the reversal the contrast is an identity rather than a test.

P3, cross-measure boundary alignment. At a genuine regime transition, several observables reorganize near the same control value. Requires a boundary located from one observable and registered before the others are examined. Without the registration the prediction is unfalsifiable in practice, because a boundary can be located in any observable after the fact.

P4, representation-first generalization. An account whose Φ is identified on one task family and whose G is identified on another predicts a third without refitting. This is Definition 4 stated as an experiment rather than as a condition.

P5, cross-architecture universality. Normalized quantities such as a margin-to-budget ratio collapse across substrates. Requires the normalization to be fixed before the second substrate is measured. Failure weakens the claim to a fundamental theory without refuting the evaluation layer.

6 Falsifiers, as conditions on the record

The programme lists seven falsifiers. Stated as conditions on a record rather than as judgments, they are checkable by a reader who did not run the experiment.

- F1. A registered budget manipulation leaves the distinction threshold and the predicted reversal classes unchanged.
- F2. A representation probe registered in advance shows no change where the account attributes a flip to representation.
- F3. Choices reverse among alternatives whose differences lie in a subspace shown to be retained, by a probe registered before the choices are collected.
- F4. A scalar model with fewer effective degrees of freedom predicts the same held-out reversals and process measures at least as well.
- F5. Registered regime boundaries fail to align across independent observables.
- F6. Cross-domain metric transfer disappears when researcher-coded encodings are replaced by independently measured ones.
- F7. No reusable relation between representational limit and behavioral transition appears across substrates.

F6 is the one the existing domain exposition already fails in its current form, because its encodings are researcher-coded. That is recorded rather than defended. Phase II exists to

answer it.

7 Alternatives that are not excluded

A foundational paper that lists no live alternative is not reporting the state of the question.

Geometry may be descriptive rather than mechanistic. A successful geometric account of endpoints does not establish that the process follows geometric paths, and the domain exposition is explicit that its basis is a modeling vocabulary. Distinguishing the two requires process evidence, which Phase III exists to collect.

Stratification may be too strong. A steep smooth family approximates most categorical transitions. The stratified model earns its place only by predicting a regularity at the boundary that the smooth family misses, and no such prediction has been registered.

Search may be the wrong abstraction. Sampling, diffusion, predictive coding and direct policy execution are all live. The account survives only if bounded search is read at a level of abstraction where some counterfactual or trajectory structure is measurable, and if nothing of that kind is measurable the dynamic layer should be reformulated rather than defended.

8 Related work

The components of this account are borrowed and the borrowing is stated here rather than in a footnote.

Finite resolution producing intransitive indifference is Luce’s semiorder [11], and the behavioral fact it models is older than the formalism [12]. Bounded resources producing satisficing rather than optimization is Simon’s [13]. The result that finite data underdetermine the model is Afriat’s on budget data [6] and Kalai, Rubinstein and Spiegler’s on choice functions [7]. Recovering an attention or information cost from choice alone is Caplin and Dean’s [8], with consideration-set identification in [9, 10].

What this paper adds is narrow. It states the vacuity of the specific four-component observer as a proposition rather than as a limitations paragraph, gives modular identification as a condition on evidence rather than as methodological advice, and organizes a phase plan by which degree of freedom each phase retires. We do not offer a better theory of choice. We offer the condition under which this one can be wrong.

9 Programme

Each phase retires one degree of freedom named in Proposition 3. That is the organizing principle, and a phase that retires none is not a phase.

Phase I is deliberately low-dimensional and is the only phase whose design is fixed at this draft. Its registration fixes the consequence vectors, the candidate retained subspaces, and the predicted ordering at each budget before any data are opened [4]. The prediction it tests is P2, because P2 is the one that a noise account does not also make.

Phase I is designed but not sealed, and the distinction is worth keeping. Its evaluator is a hosted language model pinned by alias and creation timestamp, with a preflight that refuses to run if either has moved, because the gateway publishes no revision and an alias can be repointed without the identifier changing. That converts an unverifiable provenance claim into

Table 2: The phase plan, organized by the degree of freedom each phase retires from Proposition 3. A phase that retires none is not a phase. Only Phase I has a fixed design at this draft.

	Objective	Degree of freedom retired
I	A causal observer-limited reversal under registered budget manipulation	B becomes manipulated rather than fitted
II	Representation recovered from independent tasks or learned encoders	Φ stops being researcher-coded, answering F6
III	Trajectories from process measures	π becomes measured rather than assumed
IV	A registered boundary with a prediction the smooth family misses	stratification becomes an empirical claim or is dropped
V	A collapse of normalized observables across substrates	the word fundamental acquires a referent or is withdrawn
VI	A safety-relevant boundary predicted and prevented without training on the failure	the account becomes an instrument

a check that fails loudly rather than into a sentence in a methods section. Its pass tolerances were fixed by a pilot on a seed disjoint from the run’s, at a reversal margin of 0.484 and an ambiguity tolerance of 0.05, with the rendering ceiling not binding. The pilot is spent. It grades no claim, its numbers are not revised, and the run seed is drawn only after the registration is sealed. What it has established is that the design is executable on a real evaluator and that its tolerances were not chosen with the answer in view.

10 What this paper has not done

No gate in this paper has been sealed, and Phase I is not sealable as registered. A cold reread of its registration returned three defects that change what the experiment measures, of which Definition 7 and the identity in Section 3 are two, together with five registered bars that cannot fire and an unregistered line in the sampler that decides a registered outcome [4]. The design is being repaired before it is sealed. The Phase I pilot and the instrument search are design work and grade nothing, and the one row of Table 1 they support is marked unsealed. Open 6 is unresolved and bounds every claim about partial identification. Open 8 is unresolved and bounds what a Phase I pass would be worth. No measurement of a human budget exists in this programme, and nothing here licenses carrying a language-model result across to one. The domain exposition’s measurements are retained at the class its own admission filter assigned them and are not promoted by the appearance of this paper.

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